MONASH University

MONASH INFORMATION TECHNOLOGY

P1: My Learning Plan

This task helps you connect with FIT1045/FIT1053 Introduction to Programming, its assessment, and the planning necessary to achieve different grade outcomes.

Show us that you can:

- Describe how the assessment in the unit works, including task discussion, feedback processes, tests and hurdle requirements.
- Plan out how you will engage with the unit.
- Reflect on your current knowledge and your goals in the unit.

Give us:

- A PDF with your learning plan, reflections, and signature/name indicating you understand how the unit works.

We will check that you have:

- Reviewed and selected reasonable dates for submitting your work for feedback.
- Signed or added your name to indicate you feel you have understood how the unit works.

Do the following:

1. Read this document.
2. Watch all of the videos in Moodle. They are very important content to explain to you how this unit will run and be assessed.
3. Download the Task Resources and fill in the submission template.
4. Complete your learning plan, including:
    - A description of your current knowledge of programming and computing in general.
    - A reflection on what you want to learn in this unit and why.
    - A target grade. Explain why this is your target, and what you need to do to achieve it. Relate this to the meaning of each grade described below.
    - Sign or add your name to indicate you have read and understand the unit feedback process, assessment requirements, tests, dates, and times.
    - Select your target grade in OnTrack, and update your plan. Don't worry – you can change this whenever you like, up until your final portfolio submission. This will update the target dates to keep you on track as you work towards that grade.

> **Expected Length?**
> Write as many or as few words as are needed to convey the message. Clarity and authenticity are more important than length.

5. Upload your submission document PDF to OnTrack:
    - Mark the task as **Ready for Feedback** in OnTrack and upload the required files.
6. Engage with feedback to get this signed off:

   To get the task **Complete**, you need to engage with the feedback you receive. Your tutor will review your submission and may ask you to clarify aspects of the task or include aspects you have missed.

   Remember, a **Resubmit** status is not a bad thing – your tutor is just trying to help improve your understanding. You have not lost any marks. Work with your tutor and your final portfolio will achieve a higher final score as a result! When you do resubmit, make sure to include a comment that describes what you changed. This needs to demonstrate how you have addressed all of your tutor's feedback.

> **Tip: Initiate feedback**
> When you submit your work, you can add a comment to your tutor. It is a great idea to use this to ask for specific feedback on your work. This can help make sure you receive the most effective feedback to support your learning.

Task in Detail

Programming is foundational to computer science and information technology. Through this unit, you will learn how computers work and how to get them to do what you want. Your success here will ensure that you will have the knowledge and skills to succeed throughout your future studies. This makes introductory programming the most important building block in any computing degree, and a valuable skill for any profession.

In this unit you can determine which grade you are aiming to achieve, and we will work with you to help you achieve that grade. The following sections go through the meaning of each grade, and the task and feedback timelines, to help you make an informed decision.

Unit Learning Outcomes

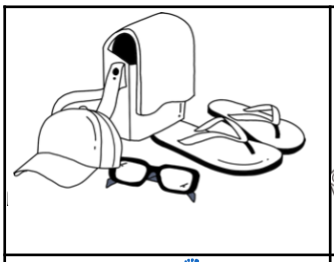
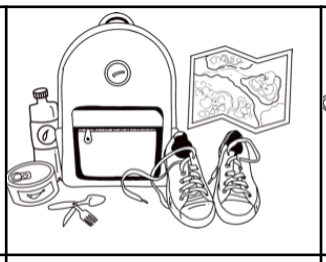
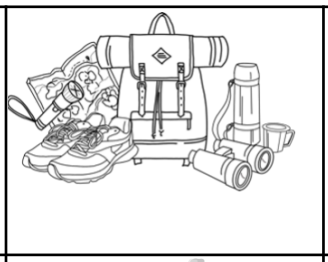
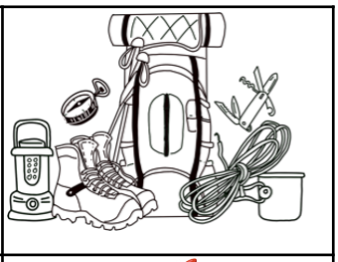
To complete this unit, you need to demonstrate that you can do the following:

- ULO 1: Effectively use variables, conditionals and loops in computer programs;
- ULO 2: Design, construct and test programs that include user-defined functions and data structures;
- ULO 3: Write programs efficiently by discovering and leveraging libraries;
- ULO 4: Implement good programming practices including unit testing, basic documentation and readability.

Through the unit you will be working on a portfolio of tasks that shows you have these capabilities. At the end of the unit you will reflect on your achievement of these Unit Learning Outcomes (ULOs) when you finalise your portfolio for submission.

Levels of achievement

Different grades require you to demonstrate different skills. The following illustrations outline the main characteristics for each grade. The visualisation uses a hiking metaphor, showing how well each grade will prepare you for future activities.

			
Competent 👣	**Capable** 🚲	**Impressive** 🚂	**Awesome** 🚀
Can work through structured resources to iteratively develop understanding. Has a foundational understanding of programming.	Can effectively utilise feedback to develop their understanding and pays attention to detail to ensure solutions work as expected. Has a good understanding of programming, and how to program.	Can work with little direction, with the persistence and adaptability needed to overcome challenges in building larger solutions. Has extended beyond programming into software development with the capability to design and build programs independently.	Can work in a proactive, self-directed manner, and demonstrates a degree of passion or inventiveness in their work. Demonstrates software development capabilities with the ability to conceptualise, design and build elegant and sophisticated programs.

When you complete all the tasks up to your target grade, you know that you have demonstrated the required skills and will receive the indicated grade at the end of the unit. You can change your target grade at any point.

The meaning of each grade is described below. It relates to the type of tasks you will be working on at that grade level, and how you should approach the unit and manage your time if targeting that grade.

1. **Pass**: Has a foundational understanding of programming; can use programming concepts.

Through the **Pass Tasks** you will develop an understanding of the basic programming tools and how to apply them to simple problems.

If you aim for a pass, you will be demonstrating that you can apply these tools and concepts to create small programs but not that you are able to create more complex programs yourself or fully understand how programs work.

If you choose to target a pass grade, you will have more time to spend on the pass tasks as you will not have to complete other tasks. However, achieving only a pass may make future computing units more challenging. Building a solid foundation here will be important for you.

Make sure you let your tutor know why you have chosen to target this grade. If the reason is that you believe you will struggle with this unit, they will support your learning as best they can to make sure you get over the line and hopefully even change your target and achieve a credit or higher.

2. **Credit**: Has a good understanding of programming, and how to program; can build things with structured guidance.

The **Credit Tasks** will ask you to think deeply about the concepts learned and how they relate to each other, so you really understand how things work. This will give you the opportunity to put into practice the concepts learned and ensure that you really know how they work together.

If you aim for credit, you will be demonstrating that you understand how programs work, and that you can draw the necessary parts together to build programs with some guidance.

If you choose to target a credit grade, you will have more time to spend on the pass and credit tasks as you will not have to complete higher-grade tasks. Make sure you let your tutor know why you have chosen to target this grade so they can provide you with the support you need.

3. **Distinction**: Has the capability to create programs independently; can build more complex things with semi-structured guidance.

The **Distinction Tasks** will ask you to create larger programs and give you the freedom to determine what these programs do. You will scope, design, code, and test these programs, working independently to create programs to meet specified requirements.

If you aim for a distinction, you will be demonstrating that you understand how programs work and can independently design and develop a program.

If you choose to target a distinction grade, you will have less time to spend on the pass and credit tasks as you will need that knowledge and time to tackle the distinction tasks. Make sure you keep

track of your time and plan ahead by preparing a timeline for yourself to ensure you can complete all the required tasks. Let your tutor know why you have chosen to target this grade so they can provide you with the support you need.

4. High Distinction: Has the capability to build elegant and sophisticated programs; can build things of your own invention.

The **High-Distinction Tasks** will ask you to demonstrate high levels of programming competency in your custom program, and your ability to incorporate concepts and tools you have learnt independently. The instructions are open ended so you have lots of freedom on what you can do.

If you aim for a high distinction, you will be demonstrating that you understand what programming can do, can independently find resources to support programming efforts, and are growing in your capability to build large, elegant, and sophisticated software solutions.

If you choose to target a high distinction grade, you will have less time to spend on the pass and credit tasks as you will need that knowledge and time to tackle the higher-grade tasks. Make sure to communicate with your tutor about your plans for the high-distinction task and discuss it with them.

5. Fail: Has not demonstrated required competency across all learning outcomes.

Failing to submit a portfolio that satisfactorily meets all of the hurdle requirements (including performance on pass tasks, tests, and task discussions/interview) will result in a fail grade being awarded. To avoid this, make sure you submit tasks regularly, work ahead of the task deadlines, respond to feedback, and attend class regularly to discuss your progress. There is plenty of feedback and support to help you pass, the main thing is to engage actively with your learning and the teaching team throughout the semester.

Feedback

Feedback provides you with information on your work and guidance on how to improve. With OnTrack, you receive feedback throughout the semester so you can improve your work and achieve the learning required.

When you submit your work in OnTrack before the due date, this will start the feedback process which you need to engage with. Feel free to be the one who initiates the feedback conversation with your tutor. Your tutor will review your work and respond with comments to indicate if there are things that you need to address or questions they have about your understanding. You may need to fix aspects of your work or discuss your understanding with the tutor before the task can be marked as Complete.

Getting the tasks marked as **Complete** in OnTrack is your way of knowing you will get the grade you are aiming for. If **all** the tasks for your target grade are Complete you already know your grade before submitting your portfolio!

You are encouraged to **submit tasks early**. This is important because you may need to take feedback in and resubmit. To get a Task marked as Complete, you need to **submit** the work ahead of the task due date, **respond** to any feedback and **resubmit** the work if required, as well as attend class to **discuss** the task and how you went with it. You might need to do this more than once.

Showing that you can receive and respond to feedback will help improve your final grade, and this process will help you learn. Ensure you make the most of the feedback dialogue and don't be scared to ask questions!

Learning with others and generative artificial intelligence

Learning with others in this unit is not only promoted but expected. Learning with peers through asking for help, supporting others, or sharing your learning has many benefits for all involved, such as social support and encouragement.

However, you need to be careful to avoid collusion when working with others!

What is collusion? Copying or letting someone copy your work. For that reason, when you help others or ask help from others, make sure the conversation focuses on explanations of how things work, how to do things, or how to think about addressing the task, rather than completed solutions.

Similar advice applies for collaborating with generative artificial intelligence (GenAI). You may choose to engage with GenAI to help your learning, but **avoid** asking it to do the programming for you. GenAI will be very capable of creating many of the simpler programs in the unit, but you need to gain these competencies yourself. If you overly rely upon GenAI you may not demonstrate the required capabilities in your task discussions or in the tests. This will result in you having to redo these tasks, slowing down your progress and putting you at risk of failing to achieve the required capabilities that will be critical to your future IT studies and to the unit.

The foundational knowledge you gain in this unit is very important, so make sure you engage in the learning, test your knowledge, and self-assessment components. There are many supports in place that can help you achieve the required capabilities, so engage with these resources when you become stuck.

Sharing of your custom programs is encouraged, but we ask you not to share your work or solutions for set activities in online sharing platforms, including GitHub.

Timelines

Teaching staff will regularly provide feedback and check tasks. We aim to provide you with feedback once per week. The following table provides suggested submission dates (based on the week of the semester). You can use this to help plan your time.

Task		Pre-reqs	Recommended Submission Week				Latest for Feedback
			👣 Pass	🚲 Credit	🚂 D	🚀 HD[+]	
P1	Learning Plan		1	1	1	1	3
P2	Building and Running	P1 👍	1	1	1	1	3
P3	Sequence	P2 👍	2	1	1	1	5
P4	Data	P3 👍	3	2	2	2	5
P5	Control Flow	P4 👍	4	4	2	2	6
P6	Structuring code	P5 👍	6	5	3	3	9
C1	Human Code	P5 👍	-	5	4	3	9
P7	Structuring data	P6 👍	7	6	5	4	9
P8	Working with multiples	P7 👍	9	8	6	5	10
T1	**Test (weeks 6-12, 14)**	P8 👍	10	9	7	6	14
C2	Sort Visualiser	P8 👍	-	9	7	6	10
P9	Member functions	P8 👍	11	10	8	7	12
C3	Generics and Operators	P9 👍	-	11	8	7	12
H1	Custom Program - Pitch	T1 ✅ C3 ✅	-	-	-	8	11
D1	Linked Lists	C3 👍	-	-	9	8	12
D2	Dynamic Array	D1 👍	-	-	10	9	12
D3	Programming Project	D2 👍	-	-	11	10	12
P10	Another Language	P9 👍	12	12	12	11	12
H2^	Custom Program	T1 ✅ H1 ✅	-	-	-	Up to you	12
H3	HD Interview Booking for Interview in week 15	T1 ✅ D3 ✅ H1 ✅	-	-	-	13 (or before)	N/A
H+1^	Something Awesome	T1 ✅ H1 ✅	-	-	-	Up to you	12
P11^	Learning Summary	P10 👍	12	12	12	12	12
H+2	Project Showcase	H3 ✅ Invite only	-	-	-	16	N/A

👍= Ready for Feedback

✅= Complete

Tasks must be submitted on or before the Portfolio submission date to be considered in your final assessment. Feedback – including marking a task as Complete – will only be given for work submitted on or before Sunday of the "latest" week. Tasks marked with a ^ can be submitted for feedback but will only be assessed in the final portfolio assessment and will never be signed off as Complete. You must submit these tasks as "I'm done" once you feel they are ready for your portfolio or they will not be considered part of your final assessment.

The Project Showcase (🎉) will be held in week 16. Students aiming for a high distinction will need to be available on campus during this week to attend this event. At the showcase, students will present a poster showcasing their work and be available to discuss/demonstrate their projects and answer questions from staff and other students. For a part of the showcase, students will be expected to view and discuss other students' projects. Participants and visitors will be asked to submit short reviews to endorse student's work as meeting the high distinction standard. Students may be asked to attend a follow up interview if reviews raise any concerns about the quality of the work or the student's ability to discuss it in a meaningful way.

Update your due dates

Note the initial due dates in OnTrack are independent of your target grade, and only reflect the **latest date** by which you should submit to get feedback. If you are aiming for a higher grade, you need to work ahead of these times to fit in the subsequent tasks.

OnTrack provides you with the ability to adjust the due dates for your tasks to align with your personal plan. Visit the plan page and choose the specific grade you are targeting. This will set your due dates to reflect the target dates for that grade.

Test

There is a test that you will need to pass to pass the unit. You will be able to sit this test in any week from week 6 onwards, although you will need to book to sit it.

When your test is assessed, it will be given one of these results:

- **Complete** – you have passed the test.
- **Redo** – you need to redo the test to demonstrate the required competencies. You will need to resit the test in a later week. If it is the final sitting, The test will be assessed as part of your portfolio and the task will not be updated in OnTrack.

You will be able to sit the test when you have completed the prerequisite tasks. The tests will be run in dedicated test sessions. The test will be conducted on paper, so remember to bring

appropriate pencil, pen, and erasers. No other material will be required or permitted. You must also be able to produce your student identification details to validate your identity.

Don't Stress

The test is only checking that you have a good understanding of the essential content. If you are progressing through the tasks, you should have no trouble completing it. You must pass this to pass the unit, but there are opportunities to re-sit if you really miss something important.

We promise that there will not be any tricky questions, there will be plenty of time, and that the practice test will be very close to the real test. Our goal is just to validate that you have achieved the required understanding to pass.

Task Discussions and Submission Limits

It is important that you are able to demonstrate programming competency as an individual. To achieve this you **must** discuss tasks in your applied class in order to get them signed off as complete.

Your task will be marked as **Discuss** in OnTrack if you submit it before the final feedback date and your submission indicates that you appear to have achieved the required competency. When a task is marked as Discuss, you must attend your applied class and discuss this task with the teaching team **within two weeks**, otherwise the task may be marked as **Assess in Portfolio**.

Your job in these task discussions is to convince the teaching team that you have developed the competencies required by the task. This should not be difficult if you have engaged with the learning resources and tackled the task yourself, and the process will also help enhance your learning if there are some aspects that you have missed or misunderstood. Where teaching staff are not clearly convinced of your competency they may ask you to do additional work, resubmit the task, and/or discuss the task again. Work with the teaching team and you will develop the level of competency you are aiming to achieve.

Tip: Make it personal!

It will be good to share challenges you have encountered, and how you overcame these in your learning journey. These personal aspects help you demonstrate that you have achieved these competencies as an individual. The teaching team will be able to relate to this – they learned to program once too!

Think of the teaching team as your personal learning coaches. We will push you to achieve the most you can.

Work submitted after the final feedback date will automatically be marked as **Assess in Portfolio**. This will mean you do not receive feedback, and that you will not have an opportunity to discuss the task with a staff member. These dates are generous, but are essential to ensure the teaching team can provide timely feedback.

While you may have some tasks that are submitted late (and marked as Time Exceeded) these should be due to exceptional circumstances. Plan to get your work in well ahead of the final dates for feedback, and this will ensure you have the time to discuss the task, address feedback, and resubmit where needed.

Resubmissions are an important part of learning in this unit. They give you the opportunity to address feedback, and to develop and demonstrate your competency. In general you will be able to resubmit each task a number of times. However, if you abuse this opportunity and are not demonstrating sufficient progress between resubmissions, staff have the option to indicate you need to complete the task on your own by marking the task as **Feedback Exceeded** in OnTrack. As with Time Exceeded, this indicated to the portfolio assessment panel that the task needs to be checked in your portfolio, and that you have not validated this competency through task discussions.

Hurdle requirements

To pass this unit you must submit a portfolio that contains:

- **All** pass tasks to a standard that demonstrates you have the required competencies.
- Test 1 marked as Complete (in weeks 6-12) or being at the required standard on the final re-sit in week 14 (status will not be updated in OnTrack).
- Sufficient tasks marked as Complete to evidence task discussion and validate the competency shown in your submitted work.

Where there is insufficient evidence of the above to convince the portfolio assessment panel that you have the required competencies, you will not be eligible to pass the unit. In the case where your work is close to demonstrating this competency, you may be invited to attend supplementary assessment. Please note that sitting supplementary assessment will limit your final result to 50P, as indicated in the supplementary assessment policy.

Final results

At the end of the semester we will assess your portfolio using a holistic rubric. This means that we look at your overall performance, including quality of your tasks, depth of your reflections, evidence of your engagement, timeliness of your submissions, and your engagement with feedback and the teaching team. This will involve two steps:

1. Determine your overall grade, based on the tasks Complete, or at the necessary standard, within your portfolio.

   This will determine your overall grade. E.g. all Pass and Credit tasks are complete, then you will get a Credit result.

2. Review the quality of your work overall, incorporating the factors mentioned above.

   This will determine the score within the grade. For example, lower quality work will result in a Low score within the grade, average quality will result in a Mid score, and high quality work will result in a High score as shown in the following table.

	Borderline	**Low**	**Mid**	**High**
Fail	Result between 0 and 40 based on number of tasks complete			
Pass	50	53	55	57
Credit	60	63	65	67
Distinction	70	73	75	77
High Distinction	80	83	85	87
High Distinction+	90	93	95	97
				100

Borderline grades may be awarded when your work is close to meeting a given grade standard.

The 100 result may be awarded in cases where work is of exceptional high quality.

Grades are awarded based on outcomes demonstrated, and are not ranked or scaled based on the performance of other students. This is not a competition.

Grades awarded for passing portfolios will be one of the numeric values shown in the table above. For example, it will not be possible to be awarded a 76 or 79.

All passing grades represent a good achievement.

How to Succeed in FIT1045/FIT1053 Introduction to Programming:

Attend classes: The applied classes give you access to tutors and other students working on the same tasks. They provide a great environment to engage with the unit, clarify difficulties and accelerate your progress. Everything you do in these classes can be included in your learning journey, helping you demonstrate what you have learnt.

Students who attend class regularly are much more likely to find the OnTrack tasks easier to complete, stay up to date, and generally find the unit manageable. Students who miss classes tend to take much longer to complete OnTrack tasks, find them more difficult to complete, and find themselves to be less knowledgeable about key topics.

Stay up to date with your tasks: plan out how you will achieve your target grade and work well ahead of the final task dates. There will be OnTrack tasks due every week, so it helps to complete tasks in a timely fashion. Leaving the tasks to the last minute runs the risk of missing feedback if you need to resubmit.

Engage and be involved: successful students are those who are actively engaged. Ask questions, be curious, be active in class and online, and reach out for help when you need it. You'll learn a lot from these interactions, so make yourself known to the teaching team and your classmates.